6 Setting up a local APT repository

If you have a plethora of machines running on a local server, which you probably have, since you are reading this guide, you will notice that it is a wasteful process of downloading packages from each machine from the internet, of course you have the alternative for syncing the `/var/apt/cache` for each machine, but that is a tedious and cumbersome process, and would not work for upgrades, instead the recommended way is to set up a local repository via the `apt-cacher`. This way, you won't download common packages more than once from official repositories. Here is the situation, we have one machine called repository-cache, this machine is going to act as the repository cache, basically, any other machines in your network is going to use it as a repository.

On your terminal, type:

```
$ sudo apt-get install apt-cacher
```

Now, we need to configure it for our specifications, since it is primarily aimed at administrators, there is no graphical user interface and all the editing will be done by modifying the `/etc/apt-cacher/apt-cacher.conf.`

So now, open apt-cacher's main configuration file: `/etc/apt-cacher/apt-cacher.conf` and start editing it according to your settings. The default port `apt-cacher` is running on is port `3142`. You might want to change this value accordingly to your needs. You won't typicaly need to bother with this unless, you have a specific service already running on that port.

`allowed_hosts:` by default, all host are allowed to use the repository cache. If you want only a particular IP range to access, you can specify here. If you want to allow over your Wi-Fi say `10.0.0.0/24` and `localhost`, then put the following values

```
allowed_hosts= 10.0.0.0/24
```

notice how localhost is always allowed by default.

Let us forget about the generate_reports option, it is also self explainatory.

Next comes the import setting of `path_map`

`path_map`: This is an interesting directive. Here you can define different aliases for different repository host. So for a typical ubuntu setup, you will have:

```
path_map =  ubuntu archive.ubuntu.com/ubuntu; ubuntu-updates archive.ubuntu.com/ubuntu ; ubuntu-security security.ubuntu.com/ubuntu
```

The mappings above are explained below:

`ubuntu` and `ubuntu-updates` to host `archive.ubuntu.com/ubuntu`